PROJECT REPORT

Auto Ticket Classification Using Flow Designer

A ServiceNow-based school IT helpdesk automation solution

Project Title	Auto Ticket Classification Using Flow Designer
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Platform	ServiceNow
GitHub Repository	Project GitHub repository

Prepared from the project activities and implementation details supplied by the team.

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This revised report does not contain screenshot placeholders. Test evidence is summarized in text and should be supplemented only if your institution separately requires screenshots.

1. INTRODUCTION

IT helpdesks receive requests involving network connectivity, hardware failures, account access and device performance. Manually categorizing each request can create repetitive work, inconsistent records and delays. This project uses ServiceNow Flow Designer to automate ticket classification through configurable rules rather than machine learning or custom scripting.

1.1 Project Overview

Auto Ticket Classification Using Flow Designer is a ServiceNow workflow designed for a school IT helpdesk. A custom table stores ticket information such as caller, ticket number, category, subcategory, short description, description, state, assigned group and assigned user. When a ticket is created, the flow evaluates its short description and assigns a matching category and subcategory for four supported issue types.

The project also configures Subcategory as a dependent choice field linked to Category and includes an email action to acknowledge ticket creation to the caller. Configuration changes are organized in an update set that can be marked Complete and exported as XML for transfer.

1.2 Purpose

The purpose is to reduce repetitive manual classification, improve consistency in ticket records and provide prompt acknowledgement to users. The project demonstrates how a low-code workflow platform can automate a common service-management process while keeping its configuration understandable and maintainable.

2. IDEATION PHASE

The ideation phase focused on common school IT helpdesk challenges and selecting a practical solution that could be implemented with ServiceNow's built-in capabilities.

2.1 Problem Statement

PS-1: School IT support staff need to categorize incoming tickets quickly, but manual classification of varying issue descriptions can create repetitive work, inconsistent records and processing delays.

PS-2: Students submitting support tickets need acknowledgement that their request has been recorded, but without an automated notification they may be uncertain whether the request was received.

2.2 Empathy Map Canvas

Says	Thinks	Does	Feels
“I need to report a Wi-Fi or hardware issue.” “I want confirmation.”	“Will my ticket be categorized correctly?” “When will I hear back?”	Submits a ticket with caller details and an issue description; checks for confirmation.	May feel uncertain without acknowledgement and wants a clear process.

This empathy map represents likely user needs for the school helpdesk scenario. It is a design aid rather than the result of formal user interviews.

2.3 Brainstorming

The team considered manual categorization, rule-based automation, dependent choice fields and email notifications. Rule-based classification was selected because the initial issue types can be identified through recognizable keywords. Flow Designer provides a visual way to implement the rules and update records without a custom classification program.

3. REQUIREMENT ANALYSIS

Requirements were derived from the stated business use case and the activities completed in ServiceNow.

3.1 Customer Journey Map

Stage	User action	System response	Expected value
1. Report issue	Caller enters ticket details and short description.	A ticket record is created in the custom table.	Structured issue capture.
2. Classify	No extra action from the caller.	Flow evaluates the short description and assigns fields for supported patterns.	Less manual categorization.
3. Confirm	Caller waits for acknowledgement.	Email action sends confirmation to the caller's email.	Improved communication.
4. Review	Support staff view the ticket.	Ticket fields are stored in a standardized structure.	Easier review and handling.

3.2 Solution Requirement

ID	Requirement	Implementation approach
FR-01	Store ticket information in a structured table.	Custom ServiceNow table with ticket fields and auto-numbering.
FR-02	Classify supported tickets automatically.	Flow Designer conditions inspect the short description.
FR-03	Populate Category and Subcategory.	Update Record actions set both fields when a condition matches.
FR-04	Limit subcategory choices by category.	Dictionary configuration makes Subcategory dependent on Category.
FR-05	Notify the caller after ticket creation.	Send Email action uses Caller → Email.
FR-06	Support configuration transfer.	Update set can be marked Complete and exported to XML.

Classification rules

Short-description keyword/pattern	Category	Subcategory
Wi-Fi or Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot password	Access	Forgot Password
Slow computer	Performance	Slow Computer

Descriptions that do not match a configured condition may remain unclassified and would require additional rules or manual review.

3.3 Data Flow Diagram

EXTERNAL ENTITY	PROCESS	DATA STORE / OUTPUT
Caller / Student Provides caller details and issue description	1. Create ticket record 2. Evaluate short description 3. Set Category and Subcategory 4. Trigger email action	School IT Ticket table stores ticket fields. Email system delivers confirmation to caller.

Figure 1. High-level data flow for ticket creation and classification.

3.4 Technology Stack

Technology / component	Purpose
ServiceNow Platform	Provides the application environment, table configuration and record forms.
Flow Designer	Runs the record-created flow, evaluates conditions and updates ticket fields.
Custom table and dictionary fields	Stores ticket information and defines field types, choices and dependencies.
Email action	Sends ticket confirmation to the caller's email address.
Local Update Sets / XML	Packages configuration changes for export and transfer.

4. PROJECT DESIGN

The design separates ticket storage, classification rules, dependent field configuration and email notification. This makes each part easier to understand and update.

4.1 Problem Solution Fit

The identified problem is repetitive manual categorization and limited immediate feedback to callers. The workflow addresses these needs by setting category fields through conditions and sending an email confirmation. It is suitable for a school helpdesk where common issue types can be identified using simple text rules.

The approach is not a machine-learning classifier. It uses explicitly configured keyword conditions, so its accuracy depends on the wording of ticket descriptions and the completeness of the rules.

4.2 Proposed Solution

A custom School IT Ticket table is configured with auto-numbering, caller reference, Category and Subcategory choices, Short Description, Description, State, Assigned Group and Assigned To. Subcategory is configured as dependent on Category.

The “Auto Classify School IT Tickets” flow runs when a record is created and Category is empty. It evaluates the short description through conditional branches. For a recognized issue, an Update Record action populates Category and Subcategory. An email action sends confirmation to the caller. The flow is activated after configuration.

4.3 Solution Architecture

Layer	Component	Responsibility
Input	ServiceNow ticket form	Accepts caller identity and ticket description.
Data	Custom School IT Ticket table	Stores ticket records and field values.
Automation	Flow Designer	Evaluates configured conditions and updates records.
Consistency	Dependent choice configuration	Aligns subcategory choices with selected category.
Notification	Send Email action	Sends confirmation to the caller.
Deployment	Update set export	Exports configuration as XML for transfer.

5. PROJECT PLANNING & SCHEDULING

The project was organized into requirements gathering, table and field configuration, workflow automation, testing and packaging. The following responsibilities reflect the task assignments supplied in the project activity descriptions. Christopher Jerome Selvan is identified as the team lead.

5.1 Project Planning and Team Responsibilities

Team member	Role	Responsibilities
Christopher Jerome Selvan	Team Lead; Flow Automation; Testing and Delivery	Led the project; implemented the Auto Classify School IT Tickets flow; configured the record-created trigger, conditional branches, Category and Subcategory updates and email action; handled the projector test scenario; completed and exported the update set as XML; prepared the project conclusion.
Siraj K	Requirements and Update Set Setup	Documented the business use case and business requirements; created the Project Update Set, set it to In Progress, selected the Global application and made it the current update set.
Anbarasan G	Custom Table and Navigation Documentation	Handled the navigation steps in picture-view activity and created the custom table for ticket records, including field structure, auto-numbering and form layout using Form Designer.
Amarnath M	Field Configuration and Data Dependency	Created the required fields with appropriate data types and configured choice values for Category, Subcategory and State; set Subcategory to depend on Category and mapped each subcategory to the relevant category.
Athesh Daniel M	Functional Testing and Email Verification	Performed the Wi-Fi test scenario using the description “WiFi not working in library” and checked the expected Network → Wi-Fi classification and caller email; documented the steps for verifying email through System Logs → Emails and Preview Mail.

The supplied activity durations did not include a full project start and end schedule. This section records task ownership and deliverables without inventing dates or completion times.

6. FUNCTIONAL AND PERFORMANCE TESTING

Functional testing checks whether configured rules produce the expected field values and whether email notifications can be inspected. The provided scenarios cover Wi-Fi and projector issues. The email verification activity checks the generated message through ServiceNow email logs.

6.1 Performance Testing

No measured response time, throughput, load-test or concurrency results were supplied. Formal performance claims cannot be made from the available evidence. A production evaluation should measure record-creation latency, flow execution time, email processing time and behaviour under concurrent submissions.

Test ID	Input / action	Expected result	Evidence status
TC-01	Create ticket: “WiFi not working in library”.	Network → Wi-Fi; email sent to caller.	Scenario documented; no separate evidence supplied.
TC-02	Search email logs by subject and open Preview Mail.	Generated email and preview can be inspected.	Procedure documented; no separate evidence supplied.
TC-03	Create ticket: “Projector not turning on”.	Hardware → Projector; email sent to caller.	Scenario documented; no separate evidence supplied.

Recommended additional tests include password-reset and slow-computer tickets, mixed capitalization, descriptions containing multiple keywords, missing caller email addresses and unmatched descriptions. These are recommendations and are not claimed as completed tests.

7. RESULTS

The project configuration covers automatic classification for four issue patterns, dependent choice behaviour, caller email notification and update set export. The recorded test scenarios specify the expected results for Wi-Fi and projector tickets, while the email verification activity defines how to inspect the generated notification.

7.1 Test Results Summary

Scenario	Expected classification	Additional expected behaviour	Status in supplied notes
Wi-Fi issue	Network → Wi-Fi	Caller receives email notification.	Test procedure and expected result documented.
Email verification	Not applicable	Email can be located by subject and inspected using Preview Mail.	Verification procedure documented.
Projector issue	Hardware → Projector	Caller receives email notification.	Test procedure and expected result documented.
Password / slow computer	Access → Forgot Password; Performance → Slow Computer	Caller receives email notification if the flow matches and executes.	Classification rules described; individual test evidence not supplied.

The table summarizes what the supplied activity notes specify. It does not represent independently audited execution logs or measured performance results.

8. ADVANTAGES & DISADVANTAGES

Advantages

• Reduces manual effort for supported ticket categories.
• Applies consistent Category and Subcategory values when rules match.
• Uses a visual low-code workflow that is easier to inspect and maintain than a custom-coded classifier.
• Sends email acknowledgement to the caller.
• Stores ticket information in a structured table.
• Update set export supports configuration transfer between ServiceNow instances.

Disadvantages and limitations

• Keyword rules may fail when users describe the same issue in unexpected terms.
• The current flow covers four defined issue patterns.
• Ambiguous descriptions may match an unintended condition or remain unclassified.
• Email delivery depends on correct caller email data and instance email configuration.
• No measured performance benchmark or full test coverage was supplied.
• The flow does not demonstrate machine learning or automated assignment to support groups.

9. CONCLUSION

Auto Ticket Classification Using Flow Designer demonstrates how ServiceNow can automate a routine school IT helpdesk workflow using built-in configuration tools. The project combines a custom ticket table, structured fields, dependent choice logic, conditional classification and email notification. It also includes test scenarios for Wi-Fi and projector issues and a method for exporting the update set as XML.

The solution addresses the goal of reducing manual categorization for supported ticket types while improving consistency and caller communication. Its current scope is rule-based and depends on configured keywords. Broader validation and performance measurements would be needed before making claims about production-scale reliability.

10. FUTURE SCOPE

• Expand the rules to cover more IT issues and common wording variations.
• Add a fallback path for unmatched descriptions so tickets can be flagged for manual review.
• Automate assignment to support groups based on category, location or severity.
• Configure Service Level Agreement (SLA) tracking and escalation rules.
• Add dashboards for ticket volume, category distribution, resolution time and unclassified tickets.
• Improve email templates with ticket number, issue summary and next-step guidance.
• Conduct systematic functional, integration, security and load testing.
• Evaluate ServiceNow predictive intelligence if more flexible text classification is required.

11. APPENDIX

GitHub repository: https://github.com/chris1206-creator/Auto-Ticket-Classification-using-Flow-Designer

Dataset link: Not applicable. The described implementation uses configured keyword conditions and ServiceNow records rather than a separate training dataset.

Source code: Omitted as requested.

Database link: Omitted as requested. Ticket records are stored in the configured ServiceNow custom table.

Project demo link: Not provided.

Configuration summary

Component	Configuration summary
Flow name	Auto Classify School IT Tickets
Trigger	Record Created; configured for records where Category is empty.
Supported classifications	Network / Wi-Fi; Hardware / Projector; Access / Forgot Password; Performance / Slow Computer.
Notification	Send Email action addressed to Caller → Email.
Update set	Project Update Set; marked Complete and exported as XML according to the activity description.

Prepared from the project description and task records supplied by the team. Details not explicitly provided are described as intended behaviour or recommendations rather than independently verified evidence.